Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Customer-focused applications engineer delivering cross-platform solutions through **requirements analysis, system integration, deployment, troubleshooting, technical documentation, and operational analytics** with measurable business value.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*
- Evaluated operational requirements and designed multi-tenant data solutions for 500+ users, reducing dashboard latency from 1.2 seconds to 350 milliseconds through schema, indexing, and access-pattern optimization at scale.
- Implemented reliable integration workflows processing 5,000+ monthly jobs with validation, reconciliation, error handling, and documented state transitions, achieving 99%+ reliability while reducing data-processing latency 35% for production operations.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*
- Translated instructor requirements into deployed AI grading workflows using React, Flask, AWS, and REST APIs, supporting 150+ users and 10,000+ daily requests with stable peak-load performance at scale.
- Troubleshot OCR, file-ingestion, model, and API failures, adding automated tests, diagnostics, fallback paths, and cost tracking while reducing customer feedback turnaround time by 60% for deployed workflows.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Contributed from requirements analysis through deployment for cross-platform data-sharing software, automating executable installation and 20+ tests across Windows, macOS, and Linux with Microsoft and GitHub stakeholders during Agile delivery.
- Resolved customer-facing API and integration issues using filtering, pagination, caching, AWS Glue, and S3, improving response times 30%, increasing pipeline efficiency 68%, and reducing technical debt 20%.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Designed eight application services for authentication, resource allocation, customer records, billing, permissions, and notifications, synchronizing two million records while aligning implementation with business workflow requirements at scale.
- Automated testing and staged deployments through Jenkins and Docker, reducing deployment time 50% while using monitoring and operational analytics to reduce resource wastage 42% and increase lead conversions 15%.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Designed the end-to-end solution workflow, mapped 12+ integration points, and presented architecture recommendations to the client before implementing cross-platform booking, payment, and real-time status capabilities at scale.
- Built and troubleshot Flask APIs, Firebase synchronization, pricing, coupons, and payment webhooks across 50,000+ records, delivering sub-200-millisecond responses and reducing manual refund handling by 70% across customer workflows.

Skills

Solutions: Requirements analysis, solution design, implementation, integration, deployment, troubleshooting
Automation: Python, Java, JavaScript, TypeScript, Shell, REST APIs, WebSockets, microservices, test automation
Platforms: AWS, Azure, Docker, Jenkins, Git, Linux, Windows, macOS, Firebase, SQL, PostgreSQL
Delivery: CI/CD, Agile, testing, monitoring, documentation, presentations, analytics, cost optimization, process improvement

Projects

HMIS - Hospital Management Information System Live | *Operational Workflows*
- Implemented consent-based patient access, appointment workflows, Twilio notifications, and Sentry monitoring, translating healthcare requirements into scalable TypeScript functionality while reducing administrative workload by 30% for operational teams.

FinNexus Live | *Financial Operations*
- Designed a fintech solution supporting 50,000+ users and one million transactions, integrating Plaid, real-time updates, advanced filtering, secure access, and containerized deployment for scalable, reliable financial operations.

Chat2SQL Code | *Cross-Platform Application Integration*
- Integrated Java and TypeScript services supporting 10+ databases, configuring Docker workflows across Windows, macOS, and Linux while implementing schema synchronization, structured validation, and streamlined local deployment for enterprise users.